East West University

Task 2 Report

Course Code: CSE475

Course Name: Machine Learning

Section: 2

Group	**Group04**
Dataset	BloodMNIST (MedMNIST v2)
Track	Track 2 + Track 3 (GCN + CNN+Att)
Proposed Direction	CNN + Attention + GCN

Submitted To,
Dr Raihan Ul Islam
Associate Professor
Department of Computer Science & Engineering
East West University

Submitted By,

Member Name	**Student ID**
Fahmida Jannat	2025-1-60-334
Nishat Subha Mithela	2023-1-60-248
Nusrat Jahan	2022-2-60-128
Humayara Nahar Moumita	2021-2-60-121

1. Introduction

This report presents the Task 2 results for the BloodMNIST project: baseline CNN classifiers, an intermediate CNN+Attention model, and our final proposed CNN+Attention+GCN architecture. Task 1 established that BloodMNIST classes show meaningful overlap in raw pixel/PCA space and that spatial regions of each image carry non-uniform information — both findings that motivate treating the image as a graph rather than relying purely on global CNN features. In Task 2, we first establish strong transfer-learning baselines, then build toward our proposed model in two stages: (1) a custom CNN with channel and spatial attention (**BloodNet**), evaluated as an ablation/intermediate step, and (2) our full proposed model, a frozen **ResNet50 backbone → attention block → residual Graph Convolutional Network (GCN)**, which converts the 7×7 ResNet50 feature map into a 49-node spatial graph and reasons over it with graph convolutions.

2. Preprocessing (applies to all models)

- All images resized to 224×224×3, RGB, batch size 32
- Undersampling applied to majority classes above a minority-based threshold (median count of the bottom half of classes), to reduce the 2.74:1 imbalance identified in Task 1
- Official MedMNIST train/val/test splits preserved (no re-splitting) to avoid leakage
- **sparse_categorical_crossentropy** loss, Adam optimizer, early stopping on validation loss

Table of original vs. final per-class training counts after undersampling.

Class ID	Class Name	Original Count	Action	Final Count	Samples Removed
0	Basophil	852	Unchanged	852	0
1	Eosinophil	2181	Undersampled	1107	1074
2	Erythroblast	1085	Unchanged	1085	0
3	Immature granulocytes	2026	Undersampled	1107	919
4	Lymphocyte	849	Unchanged	849	0
5	Monocyte	993	Unchanged	993	0
6	Neutrophil	2330	Undersampled	1107	1223
7	Platelet	1643	Undersampled	1107	536

3. Baseline Models

Four pretrained CNN backbones were fine-tuned as baselines: DenseNet121, ResNet50, EfficientNetB0, and MobileNetV3Large. Each was evaluated with the full metric set plus Grad-CAM and LIME explainability.

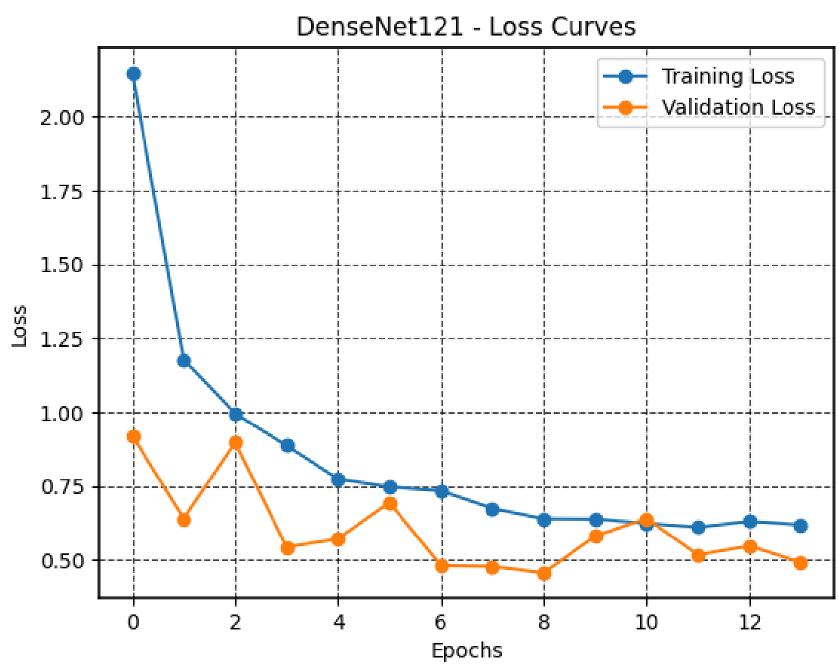
Table 1 — Baseline Model Comparison (Test Set)

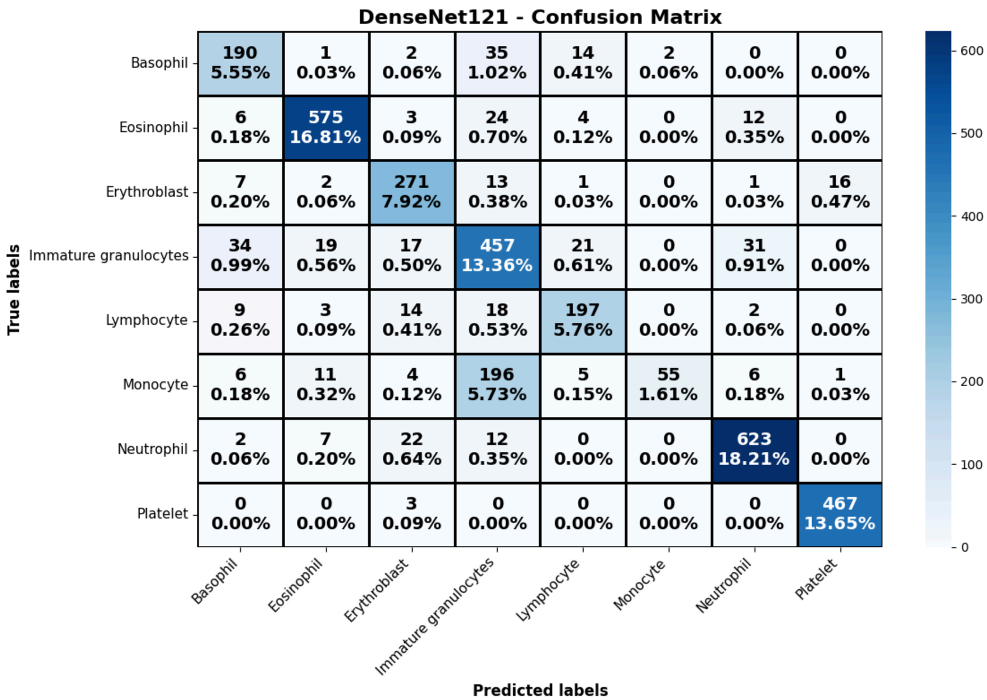
Model	Test Acc.	Macro Precision	Macro Recall	Macro F1	Avg. Specificity	Cohen's Kappa	MCC
DenseNet121	82.87%	84.46%	78.68%	78.19%	97.49%	0.8	0.8
ResNet50	90.56%	89.01%	90.38%	89.50%	98.66%	0.89	0.89
EfficientNetB0	88.92%	87.80%	88.92%	87.97%	98.43%	0.87	0.87
MobileNetV3Large	**91.26%**	**90.42%**	**90.42%**	**90.38%**	**98.74%**	**0.9**	**0.9**

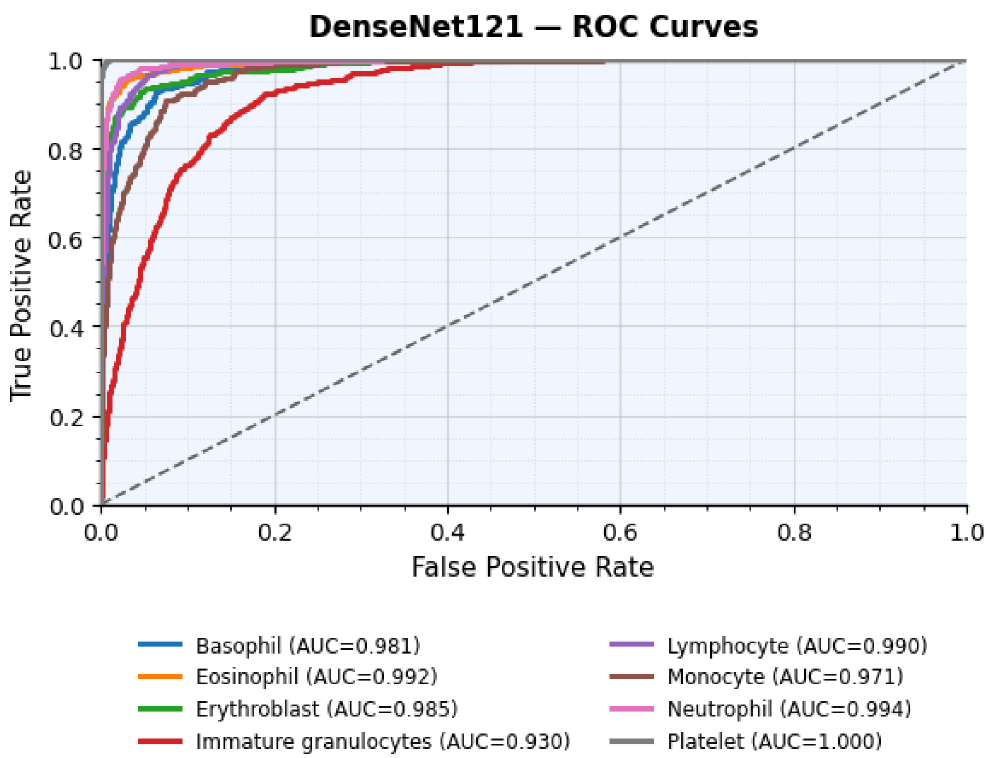
Baseline to beat: MobileNetV3Large (91.26% accuracy, 90.38% macro F1).

For DenseNet121

DenseNet121 Performance Metrics
Test Loss: 0.53
Test Accuracy: 82.87%
Macro Precision: 84.46%
Macro Recall: 78.68%
Macro F1-Score: 78.19%
Average Specificity: 97.49%
Cohen's Kappa: 0.80
Matthews Corrcoef (MCC): 0.80

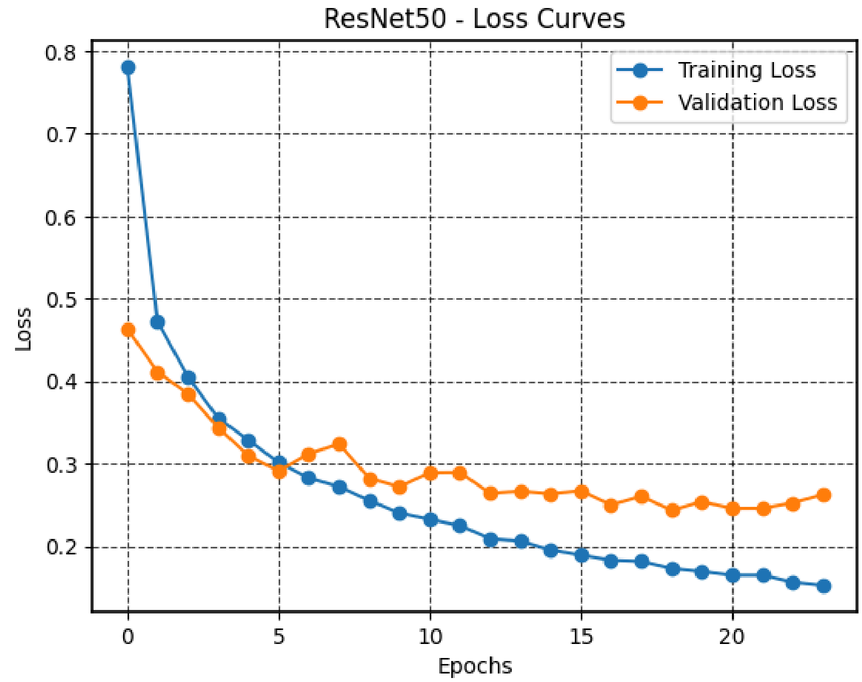


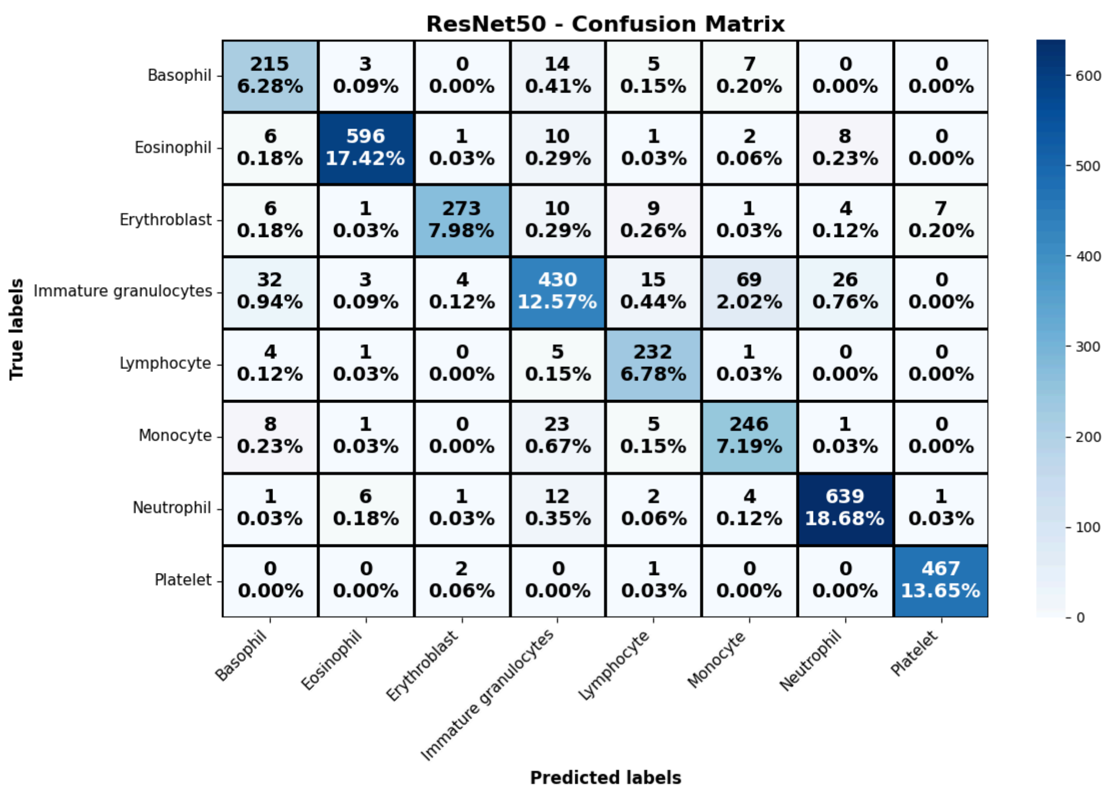

DenseNet121 - Confusion Matrix
True labels
Predicted labels
Basophil
Eosinophil
Erythroblast
Immature granulocytes
Lymphocyte
Monocyte
Neutrophil
Platelet
190 5.55% | 1 0.03% | 2 0.06% | 35 1.02% | 14 0.41% | 2 0.06% | 0 0.00% | 0 0.00%
6 0.18% | 575 16.81% | 3 0.09% | 24 0.70% | 4 0.12% | 0 0.00% | 12 0.35% | 0 0.00%
7 0.20% | 2 0.06% | 271 7.92% | 13 0.38% | 1 0.03% | 0 0.00% | 1 0.03% | 16 0.47%
34 0.99% | 19 0.56% | 17 0.50% | 457 13.36% | 21 0.61% | 0 0.00% | 31 0.91% | 0 0.00%
9 0.26% | 3 0.09% | 14 0.41% | 18 0.53% | 197 5.76% | 0 0.00% | 2 0.06% | 0 0.00%
6 0.18% | 11 0.32% | 4 0.12% | 196 5.73% | 5 0.15% | 55 1.61% | 6 0.18% | 1 0.03%
2 0.06% | 7 0.20% | 22 0.64% | 12 0.35% | 0 0.00% | 0 0.00% | 623 18.21% | 0 0.00%
0 0.00% | 0 0.00% | 3 0.09% | 0 0.00% | 0 0.00% | 0 0.00% | 0 0.00% | 467 13.65%
0
100
200
300
400
500
600

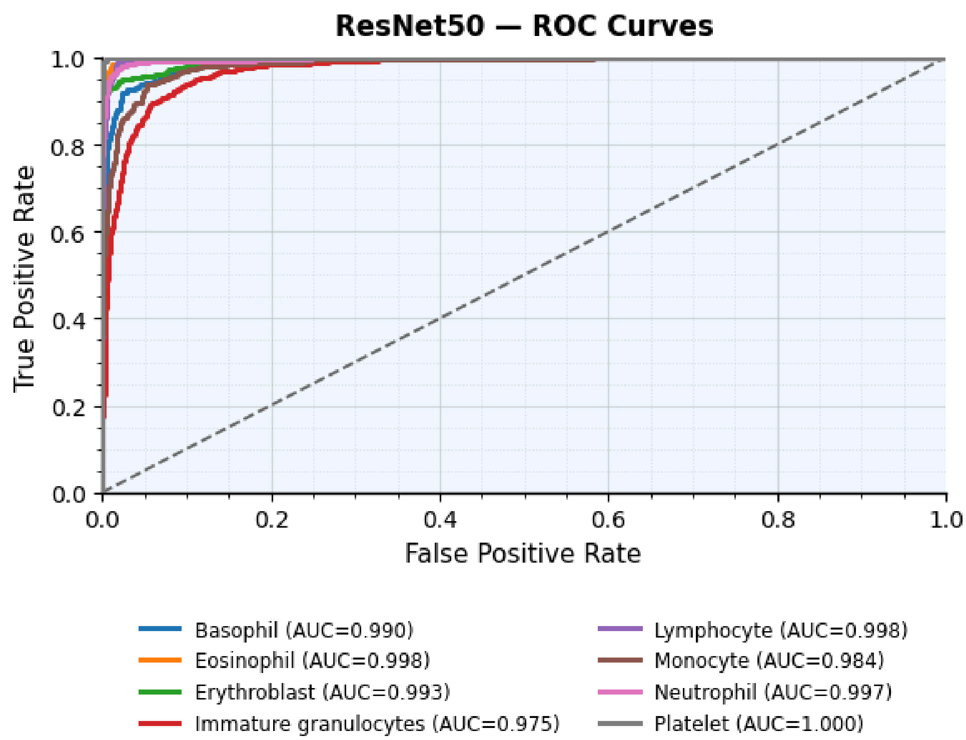

DenseNet121 — ROC Curves
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0
Basophil (AUC=0.981)
Eosinophil (AUC=0.992)
Erythroblast (AUC=0.985)
Immature granulocytes (AUC=0.930)
Lymphocyte (AUC=0.990)
Monocyte (AUC=0.971)
Neutrophil (AUC=0.994)
Platelet (AUC=1.000)

For ResNet50

ResNet50 Performance Metrics
Test Loss: 0.28
Test Accuracy: 90.56%
Macro Precision: 89.01%
Macro Recall: 90.38%
Macro F1-Score: 89.50%
Average Specificity: 98.66%
Cohen's Kappa: 0.89
Matthews Corrcoef (MCC): 0.89

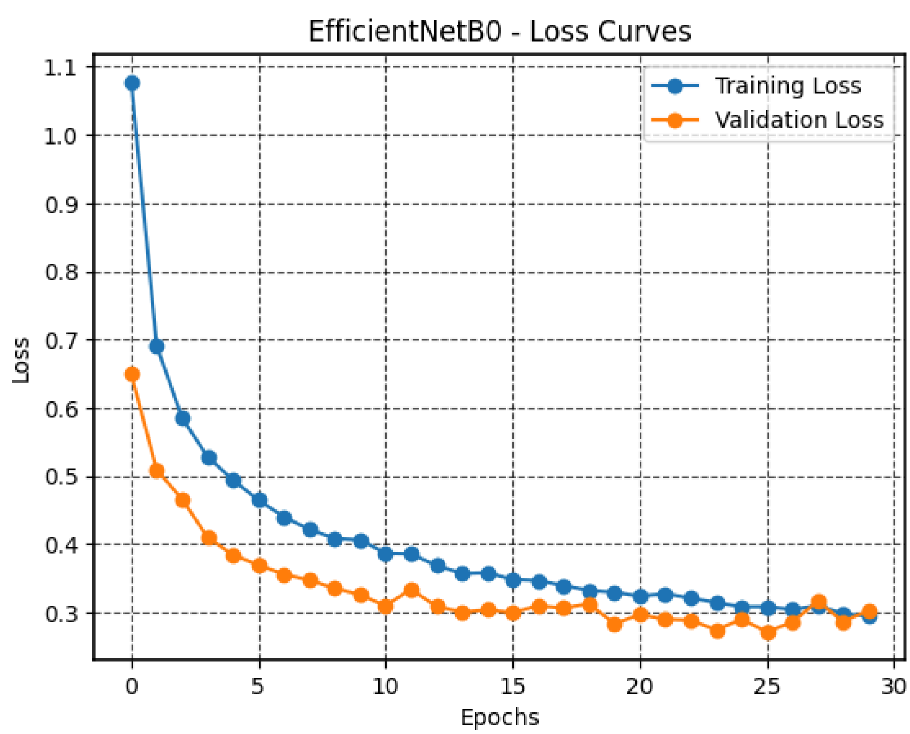


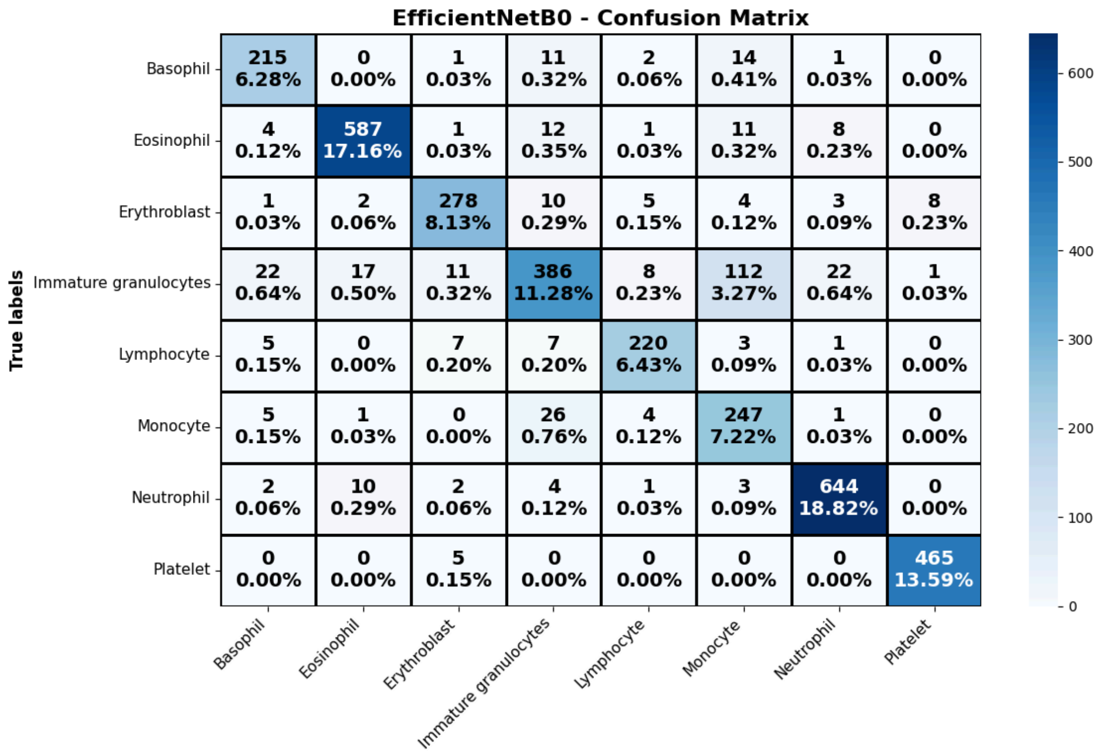


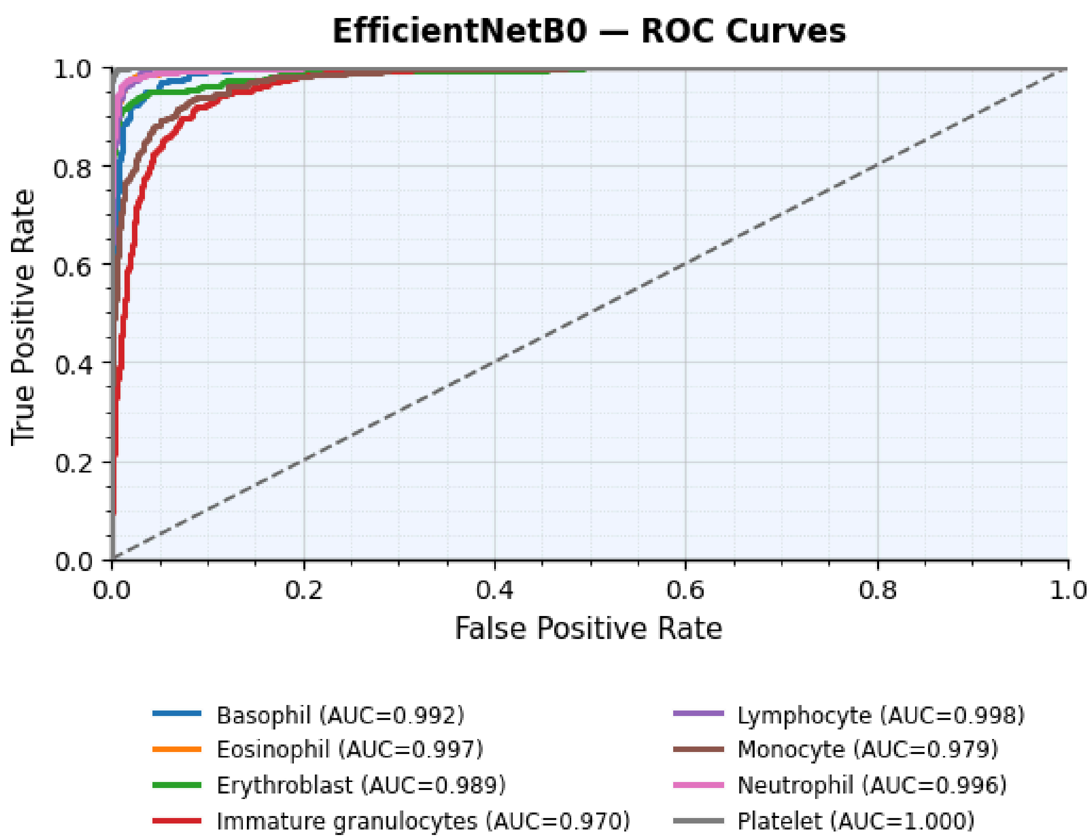


For EfficientNetB0

EfficientNetB0 Performance Metrics
Test Loss: 0.32
Test Accuracy: 88.92%
Macro Precision: 87.80%
Macro Recall: 88.92%
Macro F1-Score: 87.97%
Average Specificity: 98.43%
Cohen's Kappa: 0.87
Matthews Corrcoef (MCC): 0.87

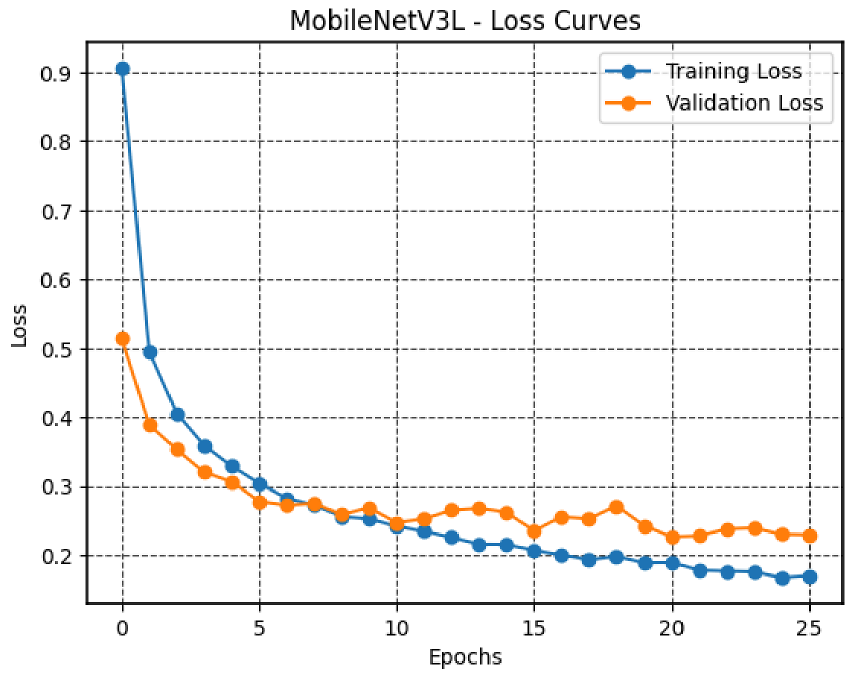


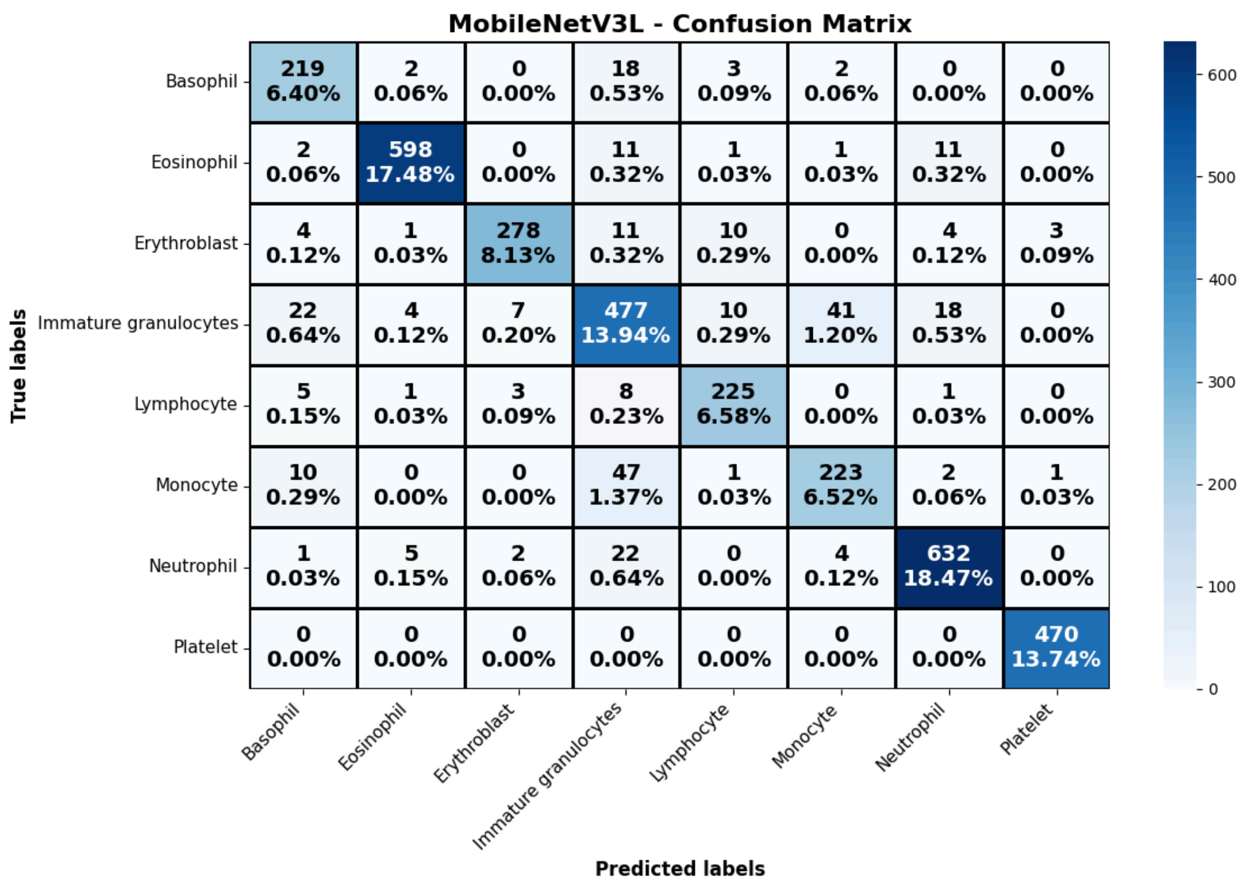

EfficientNetB0 - Confusion Matrix
True labels
Predicted labels
Basophil
Eosinophil
Erythroblast
Immature granulocytes
Lymphocyte
Monocyte
Neutrophil
Platelet
215 6.28% | 0 0.00% | 1 0.03% | 11 0.32% | 2 0.06% | 14 0.41% | 1 0.03% | 0 0.00%
4 0.12% | 587 17.16% | 1 0.03% | 12 0.35% | 1 0.03% | 11 0.32% | 8 0.23% | 0 0.00%
1 0.03% | 2 0.06% | 278 8.13% | 10 0.29% | 5 0.15% | 4 0.12% | 3 0.09% | 8 0.23%
22 0.64% | 17 0.50% | 11 0.32% | 386 11.28% | 8 0.23% | 112 3.27% | 22 0.64% | 1 0.03%
5 0.15% | 0 0.00% | 7 0.20% | 7 0.20% | 220 6.43% | 3 0.09% | 1 0.03% | 0 0.00%
5 0.15% | 1 0.03% | 0 0.00% | 26 0.76% | 4 0.12% | 247 7.22% | 1 0.03% | 0 0.00%
2 0.06% | 10 0.29% | 2 0.06% | 4 0.12% | 1 0.03% | 3 0.09% | 644 18.82% | 0 0.00%
0 0.00% | 0 0.00% | 5 0.15% | 0 0.00% | 0 0.00% | 0 0.00% | 0 0.00% | 465 13.59%

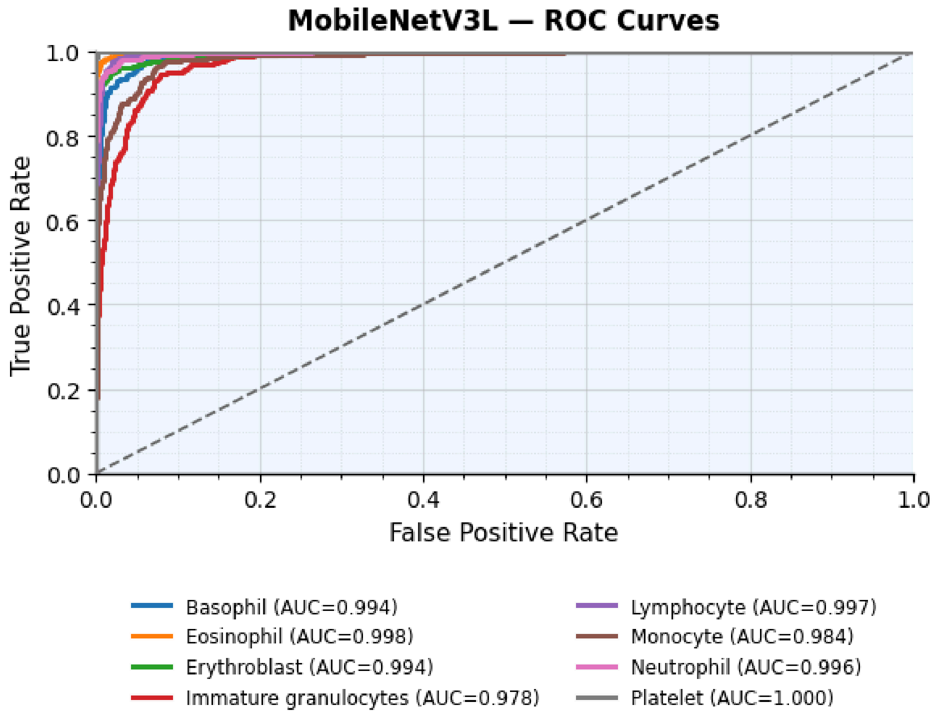

EfficientNetB0 — ROC Curves
True Positive Rate
False Positive Rate
Basophil (AUC=0.992)
Eosinophil (AUC=0.997)
Erythroblast (AUC=0.989)
Immature granulocytes (AUC=0.970)
Lymphocyte (AUC=0.998)
Monocyte (AUC=0.979)
Neutrophil (AUC=0.996)
Platelet (AUC=1.000)

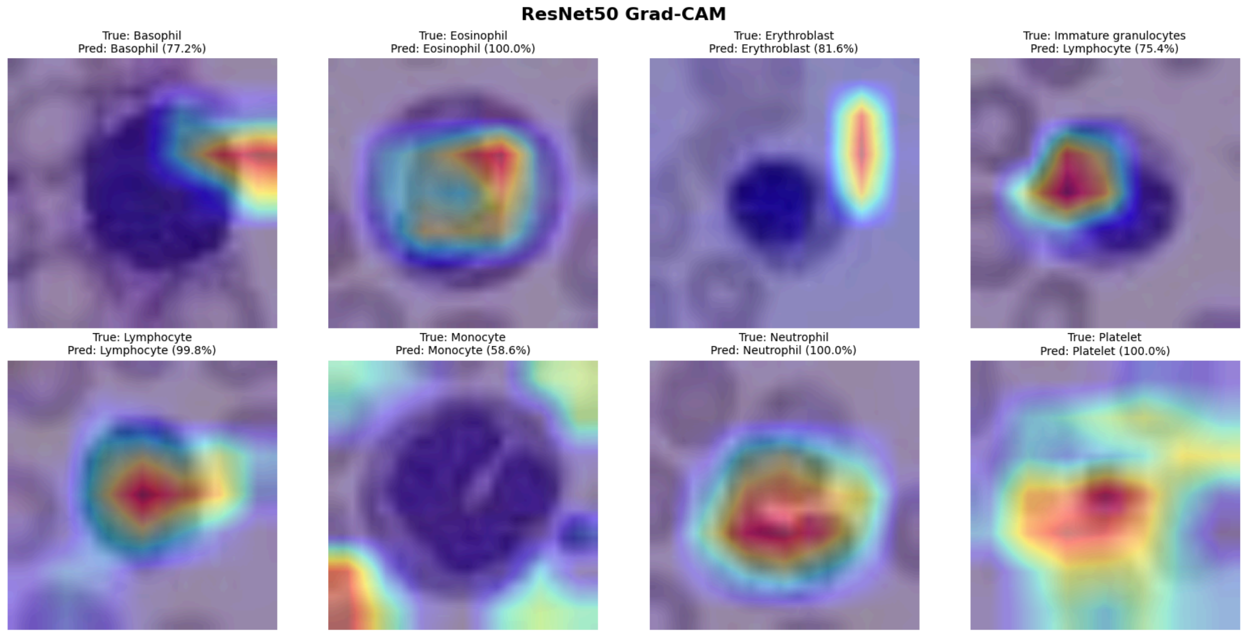
For **MobileNetV3L**

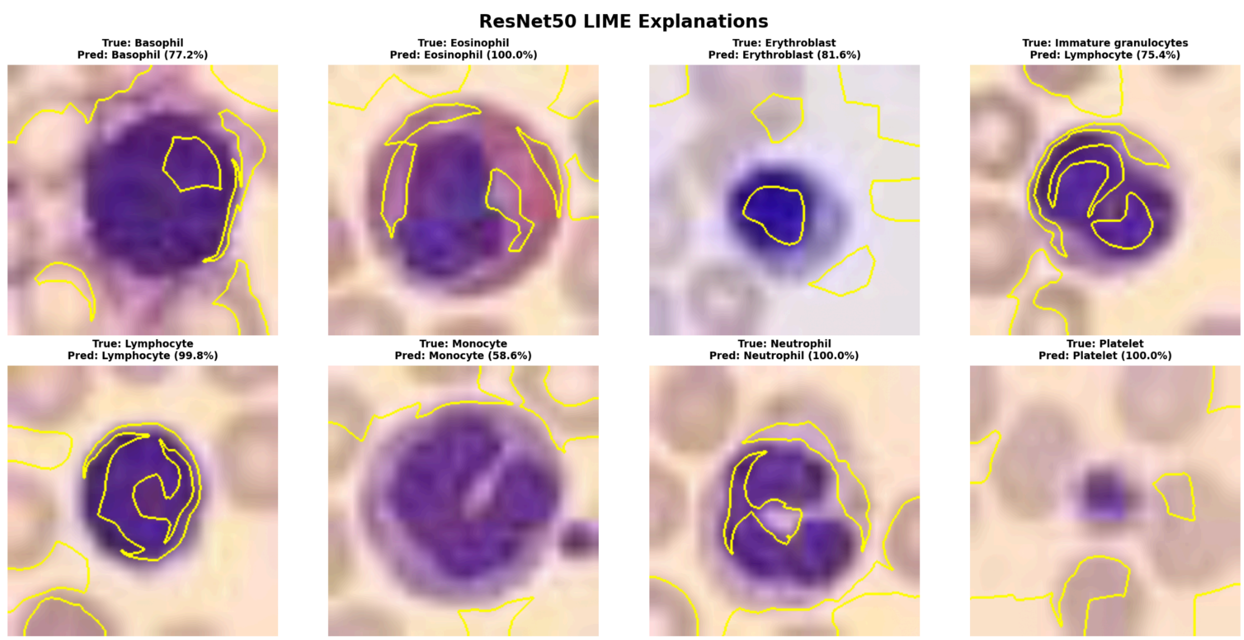
MobileNetV3L Performance Metrics
Test Loss: 0.26
Test Accuracy: 91.26%
Macro Precision: 90.42%
Macro Recall: 90.42%
Macro F1-Score: 90.38%
Average Specificity: 98.74%
Cohen's Kappa: 0.90
Matthews Corrcoef (MCC): 0.90

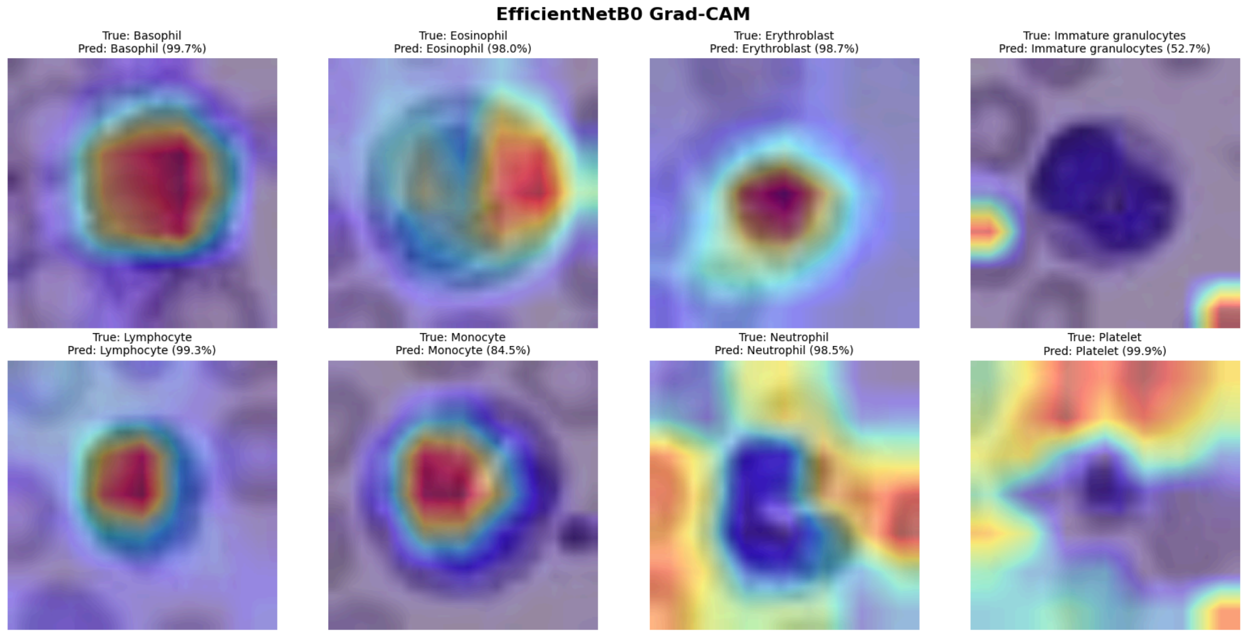


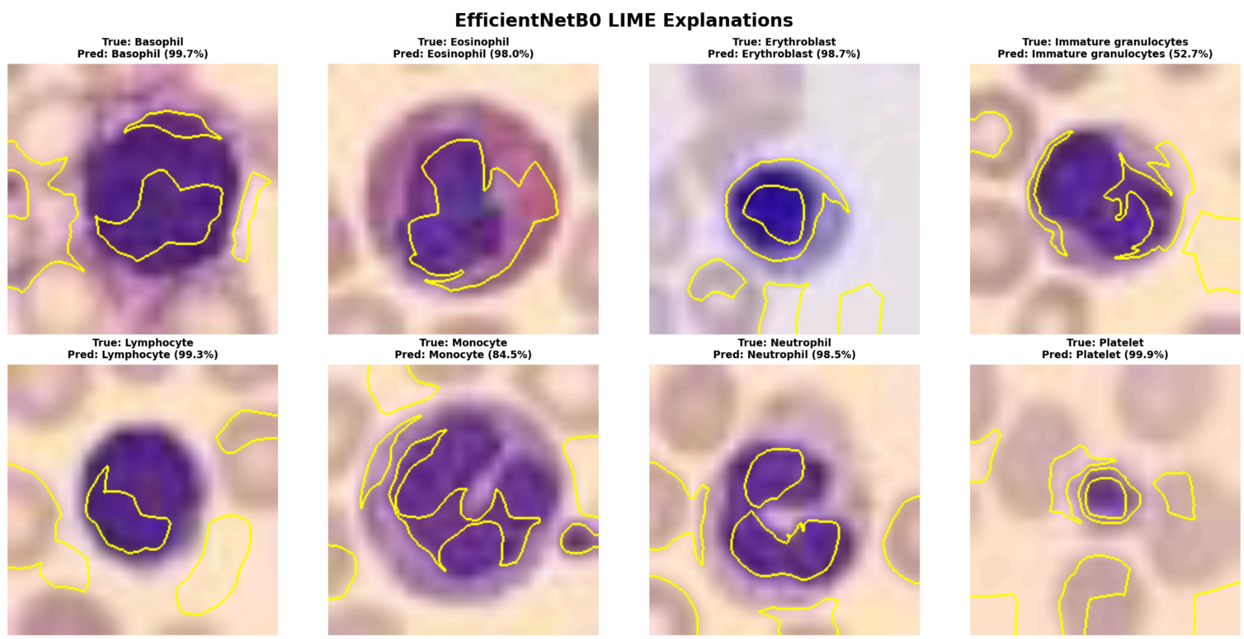


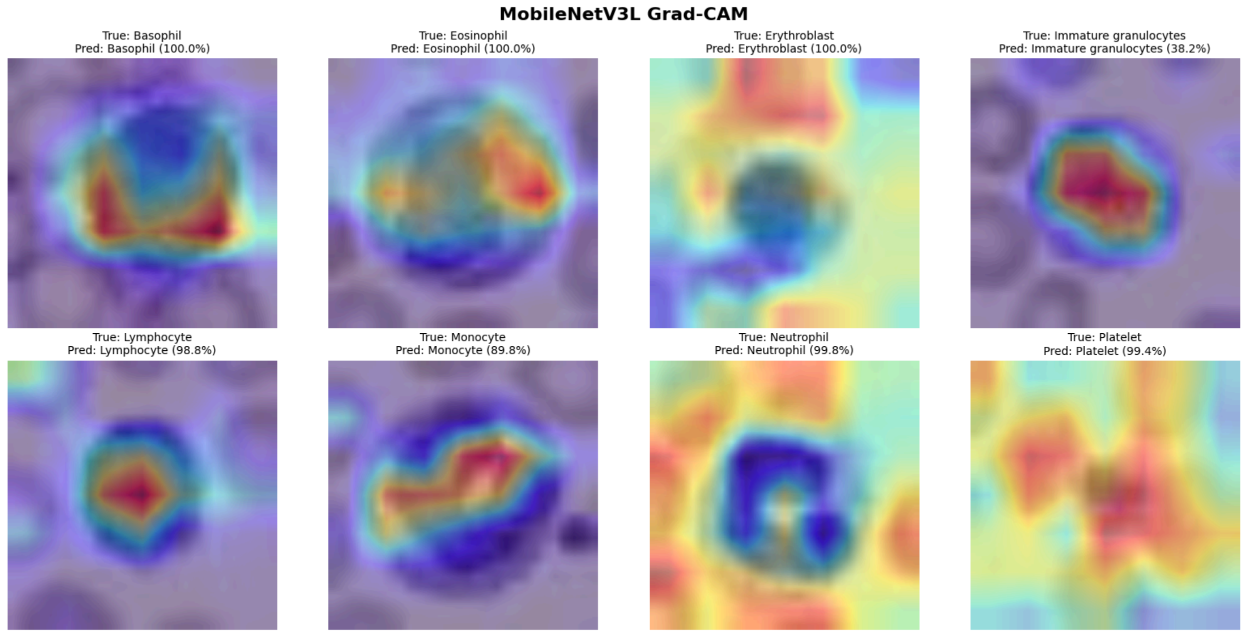

MobileNetV3L — ROC Curves
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0
Basophil (AUC=0.994)
Eosinophil (AUC=0.998)
Erythroblast (AUC=0.994)
Immature granulocytes (AUC=0.978)
Lymphocyte (AUC=0.997)
Monocyte (AUC=0.984)
Neutrophil (AUC=0.996)
Platelet (AUC=1.000)


ResNet50 Grad-CAM
True: Basophil
Pred: Basophil (77.2%)
True: Eosinophil
Pred: Eosinophil (100.0%)
True: Erythroblast
Pred: Erythroblast (81.6%)
True: Immature granulocytes
Pred: Lymphocyte (75.4%)
True: Lymphocyte
Pred: Lymphocyte (99.8%)
True: Monocyte
Pred: Monocyte (58.6%)
True: Neutrophil
Pred: Neutrophil (100.0%)
True: Platelet
Pred: Platelet (100.0%)


ResNet50 LIME Explanations
True: Basophil
Pred: Basophil (77.2%)
True: Eosinophil
Pred: Eosinophil (100.0%)
True: Erythroblast
Pred: Erythroblast (81.6%)
True: Immature granulocytes
Pred: Lymphocyte (75.4%)
True: Lymphocyte
Pred: Lymphocyte (99.8%)
True: Monocyte
Pred: Monocyte (58.6%)
True: Neutrophil
Pred: Neutrophil (100.0%)
True: Platelet
Pred: Platelet (100.0%)


EfficientNetB0 Grad-CAM
True: Basophil
Pred: Basophil (99.7%)
True: Eosinophil
Pred: Eosinophil (98.0%)
True: Erythroblast
Pred: Erythroblast (98.7%)
True: Immature granulocytes
Pred: Immature granulocytes (52.7%)
True: Lymphocyte
Pred: Lymphocyte (99.3%)
True: Monocyte
Pred: Monocyte (84.5%)
True: Neutrophil
Pred: Neutrophil (98.5%)
True: Platelet
Pred: Platelet (99.9%)


EfficientNetB0 LIME Explanations
True: Basophil
Pred: Basophil (99.7%)
True: Eosinophil
Pred: Eosinophil (98.0%)
True: Erythroblast
Pred: Erythroblast (98.7%)
True: Immature granulocytes
Pred: Immature granulocytes (52.7%)
True: Lymphocyte
Pred: Lymphocyte (99.3%)
True: Monocyte
Pred: Monocyte (84.5%)
True: Neutrophil
Pred: Neutrophil (98.5%)
True: Platelet
Pred: Platelet (99.9%)


MobileNetV3L Grad-CAM
True: Basophil
Pred: Basophil (100.0%)
True: Eosinophil
Pred: Eosinophil (100.0%)
True: Erythroblast
Pred: Erythroblast (100.0%)
True: Immature granulocytes
Pred: Immature granulocytes (38.2%)
True: Lymphocyte
Pred: Lymphocyte (98.8%)
True: Monocyte
Pred: Monocyte (89.8%)
True: Neutrophil
Pred: Neutrophil (99.8%)
True: Platelet
Pred: Platelet (99.4%)

4. Proposed Model — Stage 1: BloodNet (CNN + Attention)

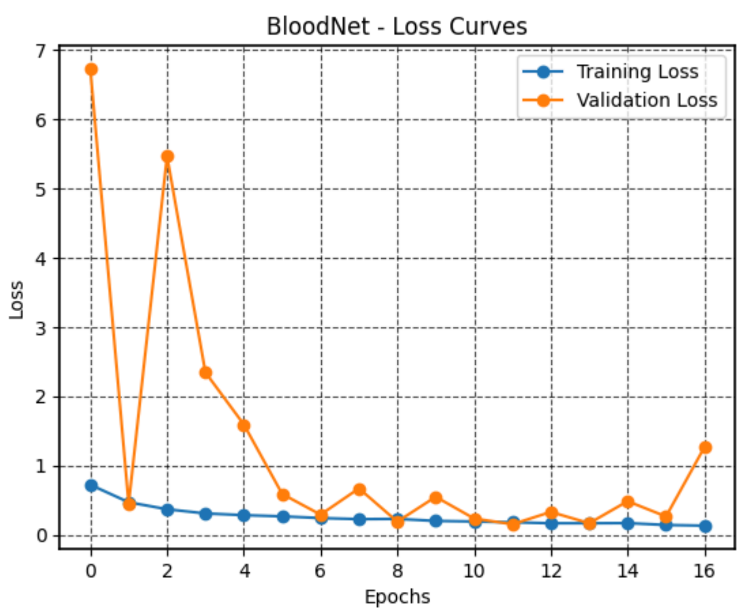
As an intermediate step toward the full proposed model, we built a custom CNN, **BloodNet**, using stacked convolutional blocks each followed by a combined channel-and-spatial attention module (in the style of CBAM):

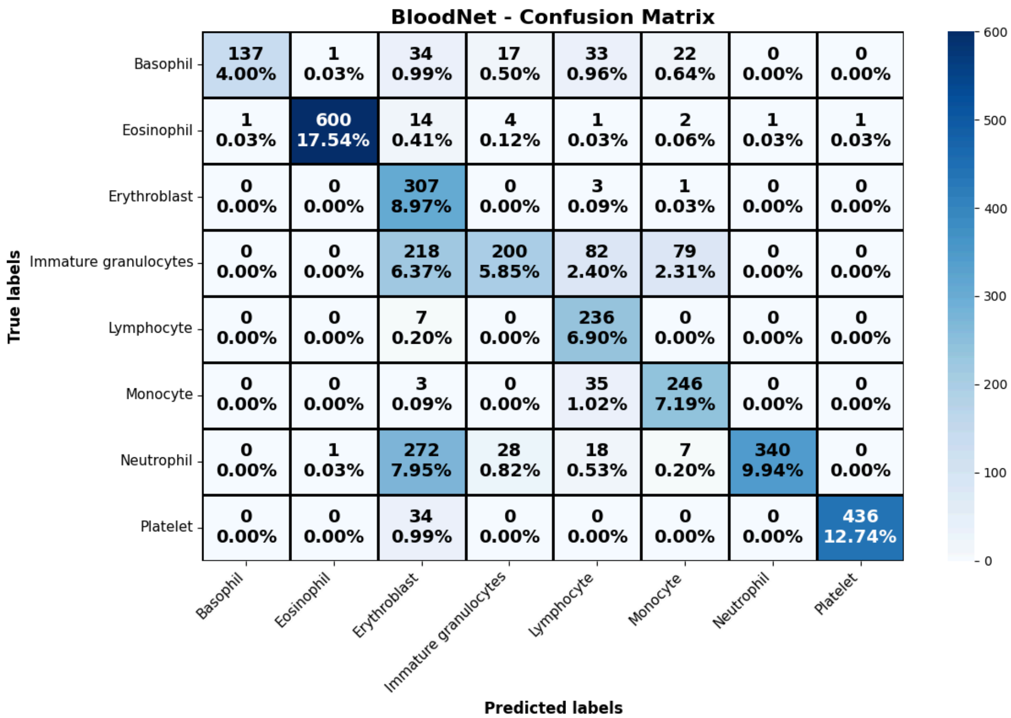
- **Channel attention**: global average-pooled and global max-pooled features passed through a shared two-layer MLP (reduction ratio 8), summed, and passed through a sigmoid to reweight channels
- **Spatial attention**: a 7×7 convolution over the channel-attended feature map, sigmoid-activated, to reweight spatial locations
- Trained for up to 30 epochs with early stopping (patience 5); stopped at epoch 17

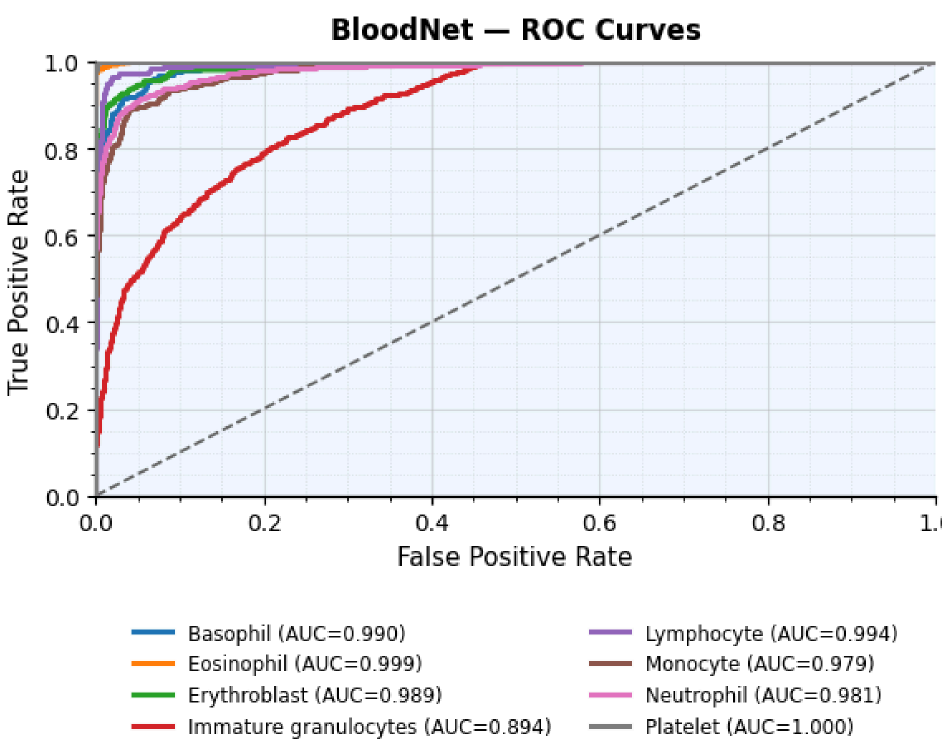
Table 2 — BloodNet Result

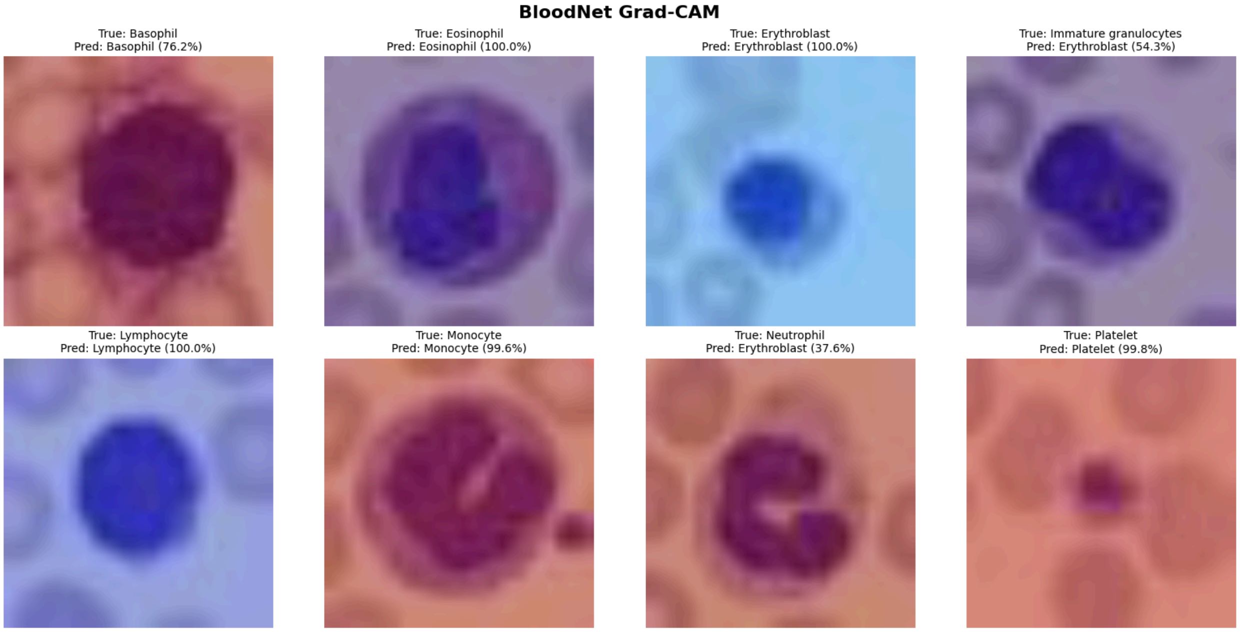
Model	Test Acc.	Macro Precision	Macro Recall	Macro F1	Avg. Specificity	Cohen's Kappa	MCC
BloodNet (CNN+Attention)	73.14%	80.00%	76.64%	72.75%	96.30%	0.69	0.71

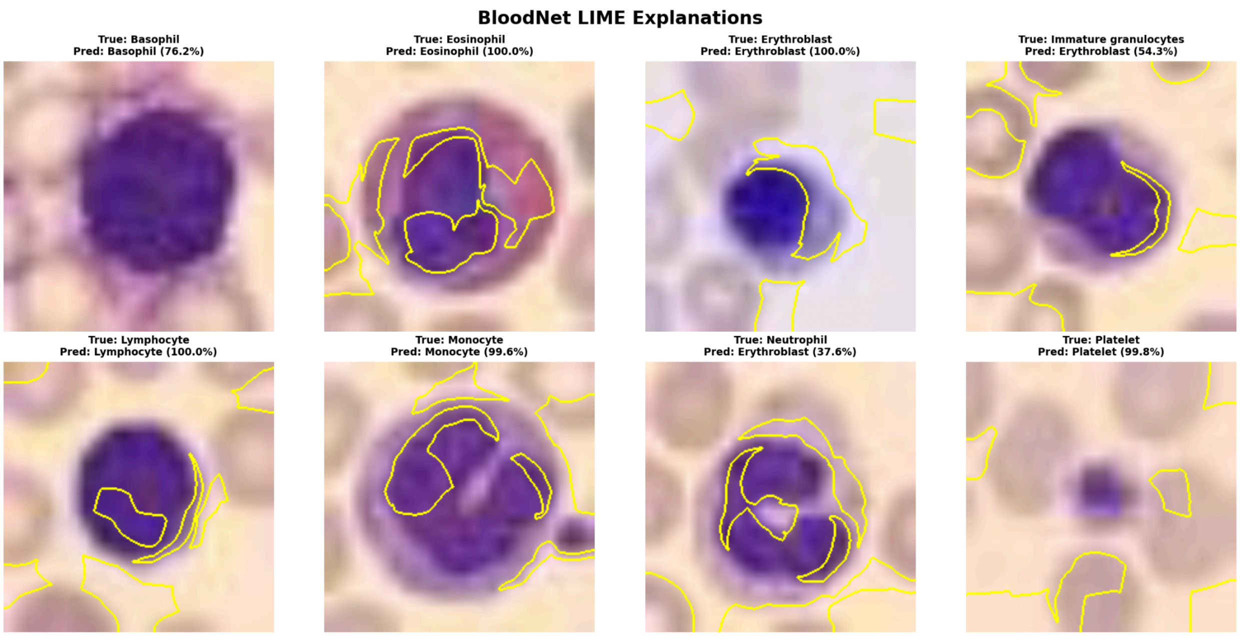
BloodNet underperforms every baseline. The training curve shows unstable validation accuracy across epochs (e.g., dropping from ~94% to ~75% between consecutive epochs before early stopping triggered), indicating the custom architecture — trained from scratch without pretrained weights — struggles to generalize on this dataset size. This result is reported as an honest ablation: it shows that attention alone, without a pretrained backbone or a relational/graph mechanism, is not sufficient to beat transfer-learning baselines on BloodMNIST.




BloodNet - Confusion Matrix
True labels
Predicted labels
Basophil
Eosinophil
Erythroblast
Immature granulocytes
Lymphocyte
Monocyte
Neutrophil
Platelet

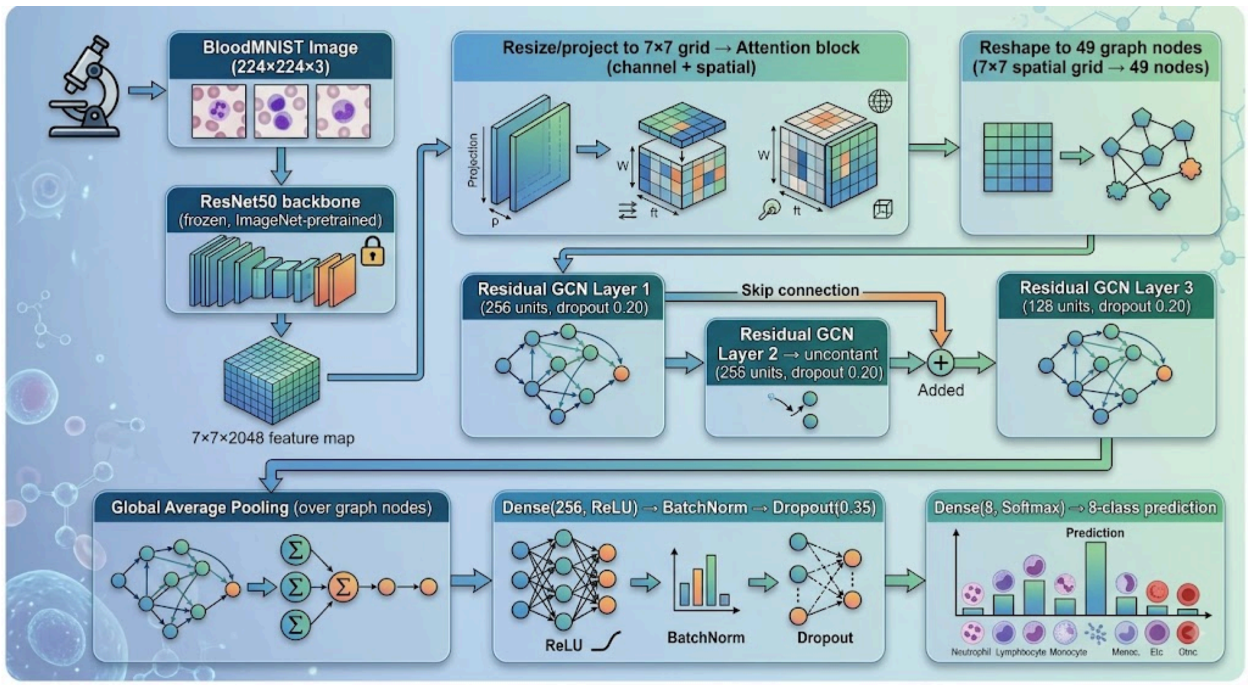

BloodNet — ROC Curves
True Positive Rate
False Positive Rate
Basophil (AUC=0.990)
Eosinophil (AUC=0.999)
Erythroblast (AUC=0.989)
Immature granulocytes (AUC=0.894)
Lymphocyte (AUC=0.994)
Monocyte (AUC=0.979)
Neutrophil (AUC=0.981)
Platelet (AUC=1.000)


BloodNet Grad-CAM
True: Basophil
Pred: Basophil (76.2%)
True: Eosinophil
Pred: Eosinophil (100.0%)
True: Erythroblast
Pred: Erythroblast (100.0%)
True: Immature granulocytes
Pred: Erythroblast (54.3%)
True: Lymphocyte
Pred: Lymphocyte (100.0%)
True: Monocyte
Pred: Monocyte (99.6%)
True: Neutrophil
Pred: Erythroblast (37.6%)
True: Platelet
Pred: Platelet (99.8%)


BloodNet LIME Explanations
True: Basophil
Pred: Basophil (76.2%)
True: Eosinophil
Pred: Eosinophil (100.0%)
True: Erythroblast
Pred: Erythroblast (100.0%)
True: Immature granulocytes
Pred: Erythroblast (54.3%)
True: Lymphocyte
Pred: Lymphocyte (100.0%)
True: Monocyte
Pred: Monocyte (99.6%)
True: Neutrophil
Pred: Erythroblast (37.6%)
True: Platelet
Pred: Platelet (99.8%)

5. Proposed Model — Stage 2: ResNet50 + Attention + Residual GCN (Final Proposed Model)

5.1 Architecture



5.2 Graph construction

- **Nodes**: each of the 49 spatial locations in the 7×7 ResNet50 feature map (after attention) becomes one graph node; node features = the attention-weighted channel vector at that location
- **Edges**: a fixed 4-neighbor grid adjacency (up/down/left/right), normalized — 217 non-zero entries across 49 nodes
- **Why this fits**: Task 1's patch-level spatial variability analysis showed that visual information (intensity, texture) is not uniformly distributed across the image — center vs. border patches differ meaningfully. A grid graph over CNN feature-map locations lets the model explicitly propagate and combine information between spatially adjacent regions, which a plain CNN's local convolutions and pooling do only implicitly.
- **Graph built after the ResNet50 forward pass on already-split data** — no leakage, since the backbone is frozen ImageNet weights (not fit on this dataset) and the graph structure (fixed grid adjacency) doesn't depend on any data statistics at all.

5.3 Explanation in plain words

Think of the image as being cut into a 7×7 checkerboard of "regions" after the ResNet50 backbone extracts high-level features for each region. Instead of just averaging all 49 regions together (as a normal CNN classifier head would), the GCN lets each region "talk" to its immediate neighbors over 3 rounds of message passing, updating its own representation based on what nearby regions look like. The residual (skip) connections between GCN layers help preserve information from earlier layers so it isn't lost as the graph reasoning gets deeper. This should help distinguish visually similar blood cell classes (e.g., Basophil vs. Neutrophil) that differ mainly in localized structural details (granule patterns, nucleus shape) rather than global color/texture — information a purely global CNN feature vector can blur together.

5.4 Training

- Backbone frozen (only attention block + GCN layers + classifier head trained)
- Class weights applied to counter residual imbalance
- Adam optimizer (lr=0.001), early stopping (patience 6, restore best weights) + ReduceLROnPlateau
- Trained up to 30 epochs

5.5 First Results

Table 3 — Final Comparison (Baselines vs. Proposed Model)

Model	Test Acc.	Macro Precision	Macro Recall	Macro F1	Cohen's Kappa	MCC
Best baseline: MobileNetV3Large	91.26%	90.42%	90.42%	90.38%	0.9	0.9
BloodNet (CNN+Attention, ablation)	73.14%	80.00%	76.64%	72.75%	0.69	0.71
ResNet50+Attention+GCN (proposed)	**91.70%**	**90.33%**	**91.52%**	**90.87%**	**0.9032**	**0.9034**
ResNet50+Attention+GCN + D4 TTA	**94.30%**	**93.32%**	**94.02%**	**93.62%**	—	—

5.6 Interpretation

The proposed ResNet50+Attention+GCN model beats the strongest baseline (MobileNetV3Large) on accuracy, macro F1, macro recall, Cohen's Kappa, and MCC, with only a marginal drop in macro precision (90.33% vs. 90.42%) — a genuine, if modest, improvement without any test-time tricks. Applying D4 test-time augmentation (8-way flip/rotation ensembling) pushes performance further to 94.30% accuracy / 93.62% macro F1, a clear margin over every baseline. This supports the Task 1 research gap: adding graph-based spatial reasoning on top of CNN+attention features provides real benefit beyond what transfer-learning CNN baselines achieve alone, while the BloodNet ablation confirms that attention by itself (without the pretrained backbone or the graph) is not enough to reach baseline-level performance.

6. Conclusion

Task 2 establishes MobileNetV3Large (91.26% acc / 90.38% F1) as the baseline to beat. Our proposed ResNet50+Attention+Residual GCN model surpasses it (91.70% acc / 90.87% F1, rising to 94.30%/93.62% with D4 TTA), while the standalone BloodNet CNN+Attention model underperforms all baselines — an honest result showing that the graph-reasoning component, not attention alone, drives the improvement. Task 3 will focus on ablating the GCN's individual design choices (depth, edge structure, attention placement), running 5-fold cross-validation with significance testing against MobileNetV3Large, and applying Grad-CAM/LIME/node-importance explainability to the final model.